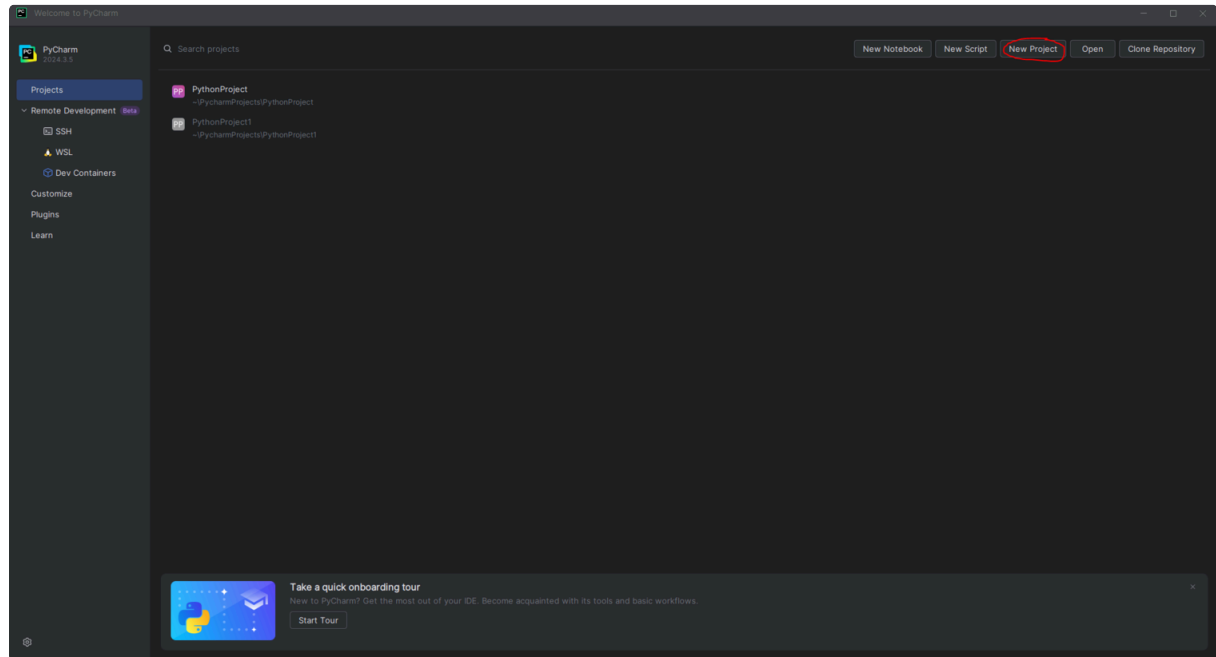


Coursework Q&A

PyCharm Configuration

1. When you open PyCharm, you could start by clicking "New Project"



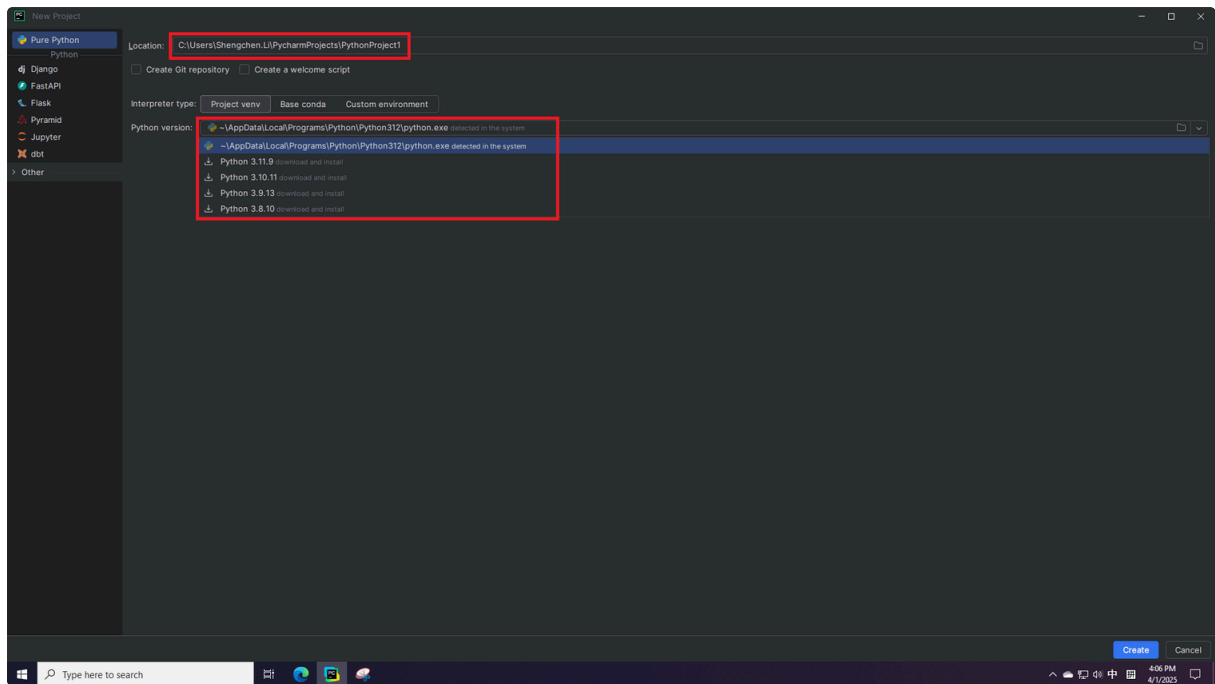
2. There are two parameters that you need to pay attention to:

- The location of the project

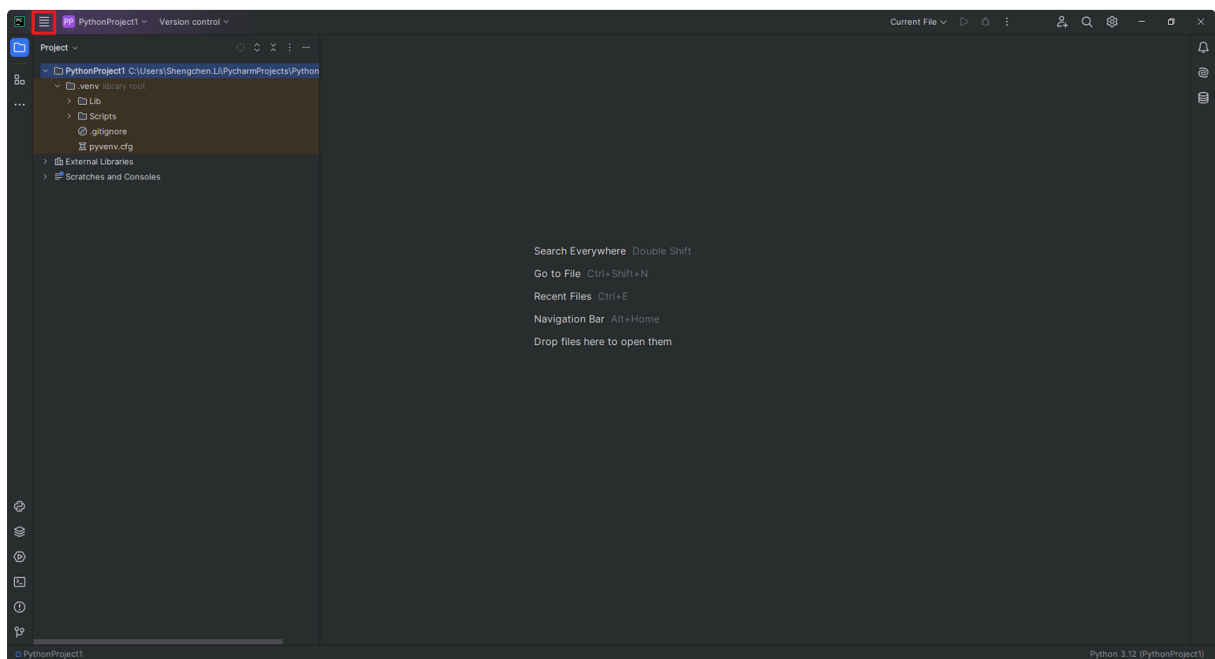


Over the live demonstration session, you need to make sure that the path of the location is as required.

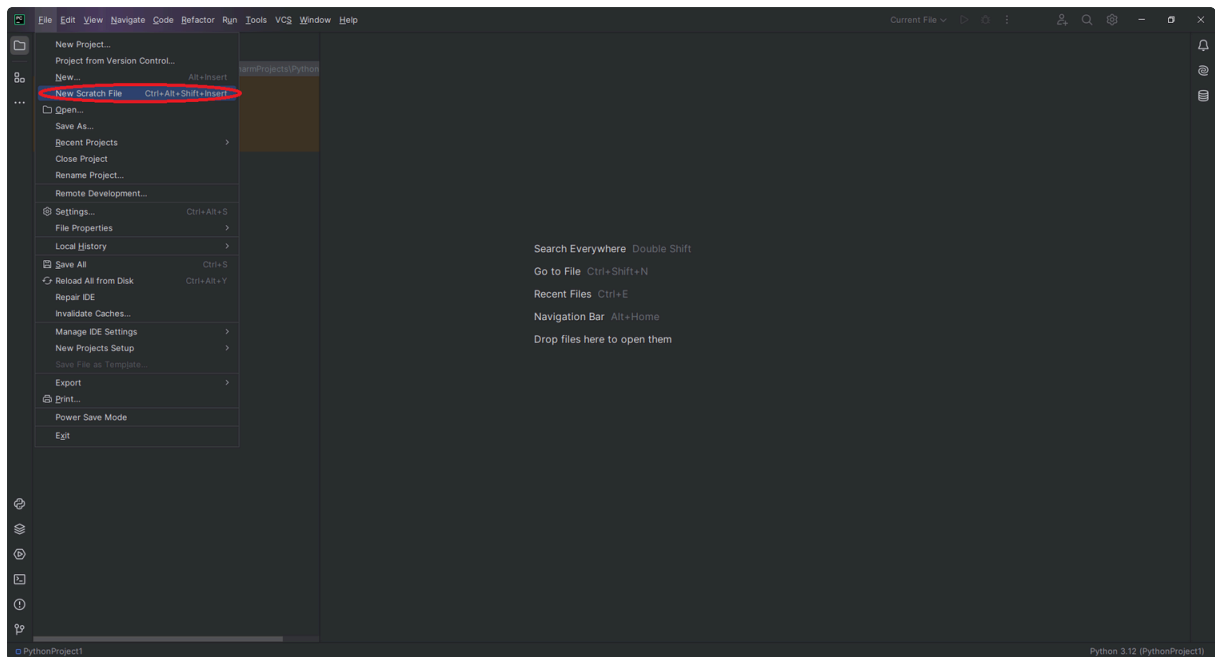
- The Python interpreter can be selected at "Python Version". If there is an existing Python interpreter, you can use that. You may also use other Python interpreters that are labelled "download and install" aside, which would give you a clean installation.



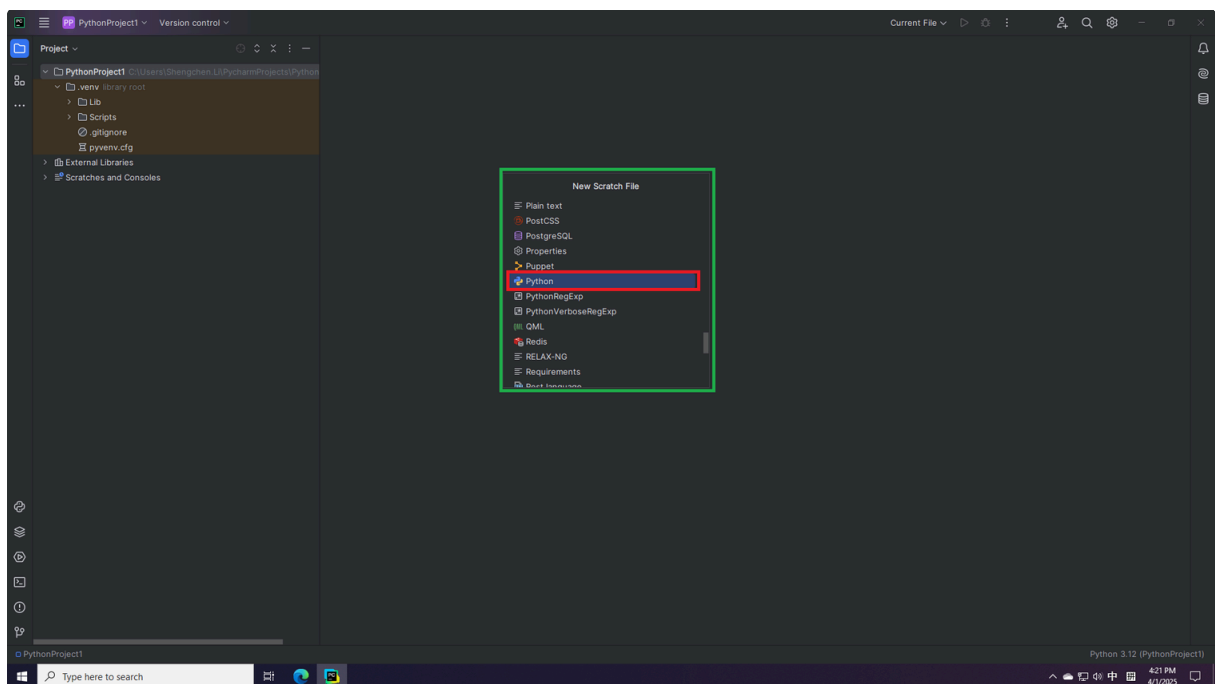
3. Click the "menu" button



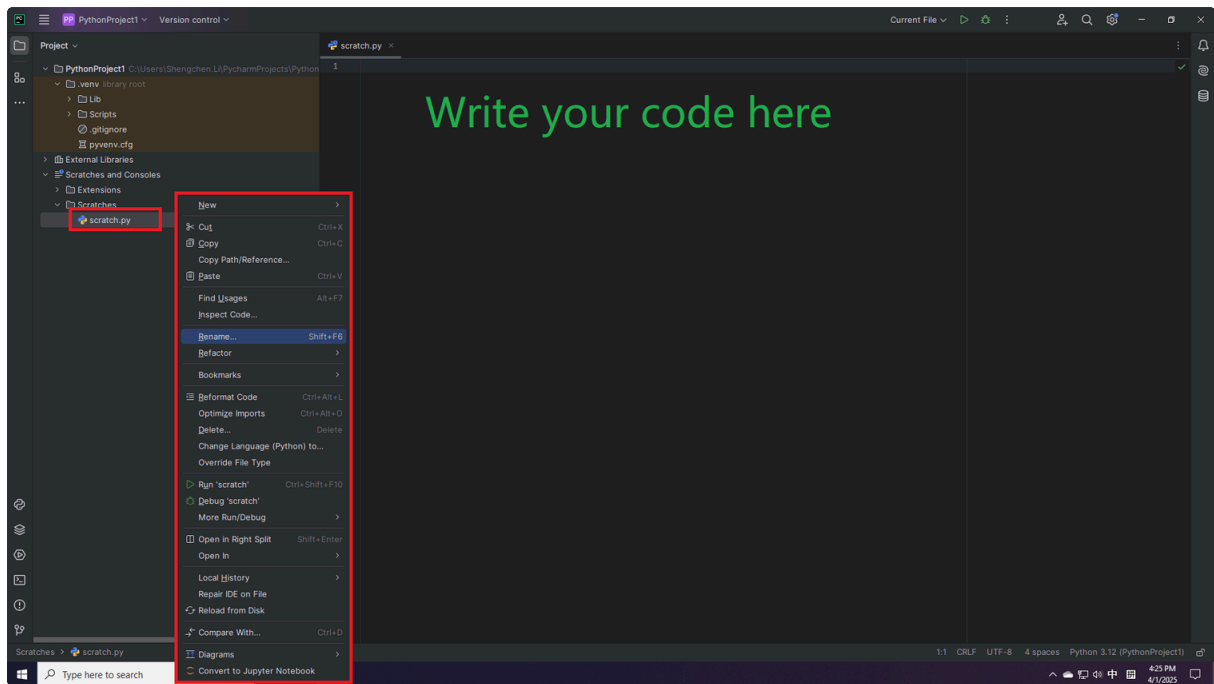
4. Click "New Scratch File"



5. In the box of scratch file, find "Python"



6. You may then type in your code in the interface. If you want to rename your file, right click the file and then select "rename".

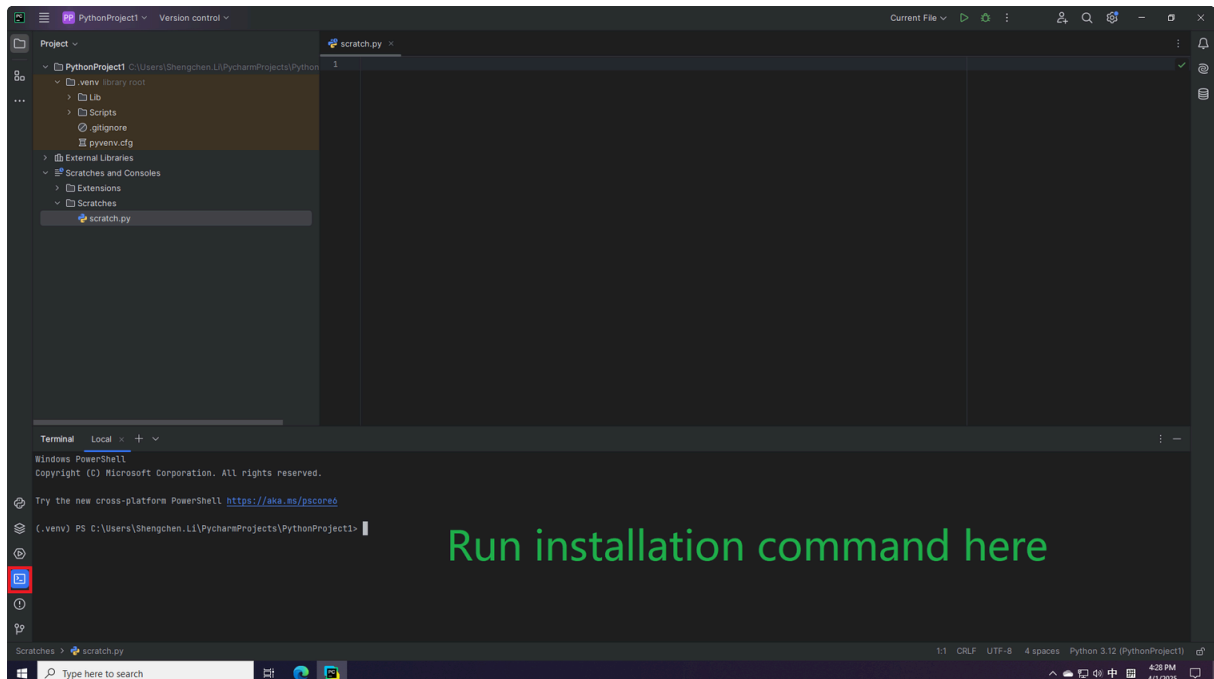


- When necessary, click "Terminal" button and install the package you need by

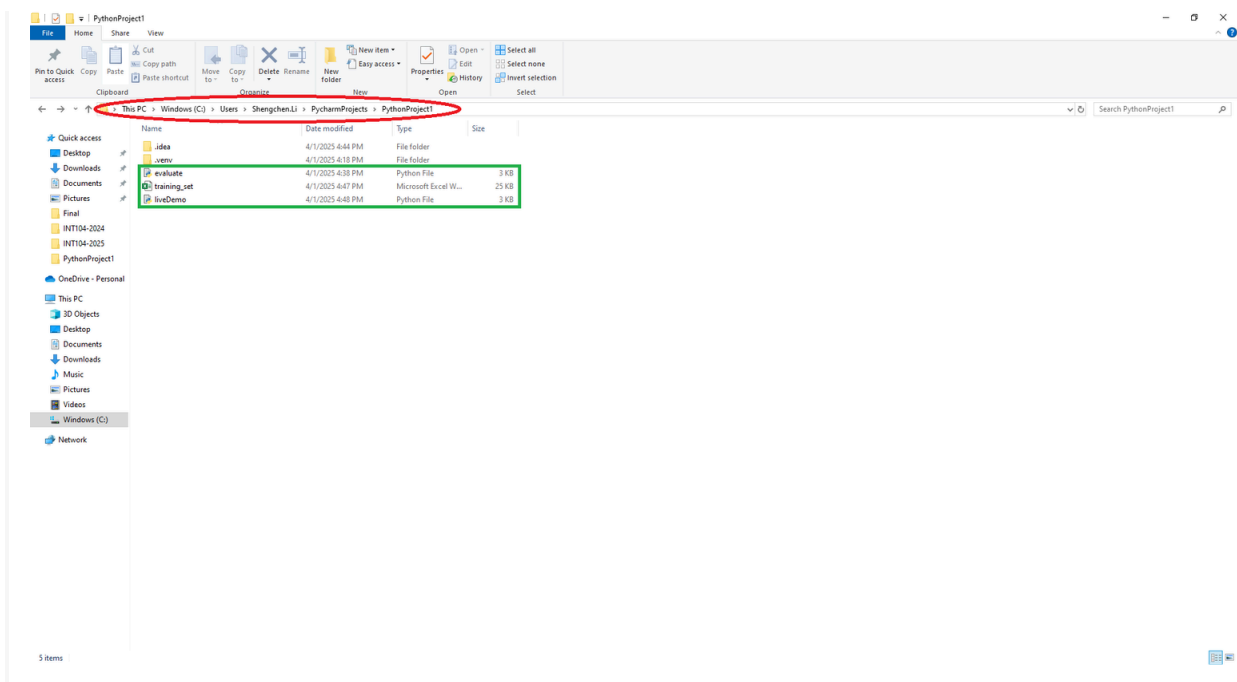
Python

```
pip install xxxx
```

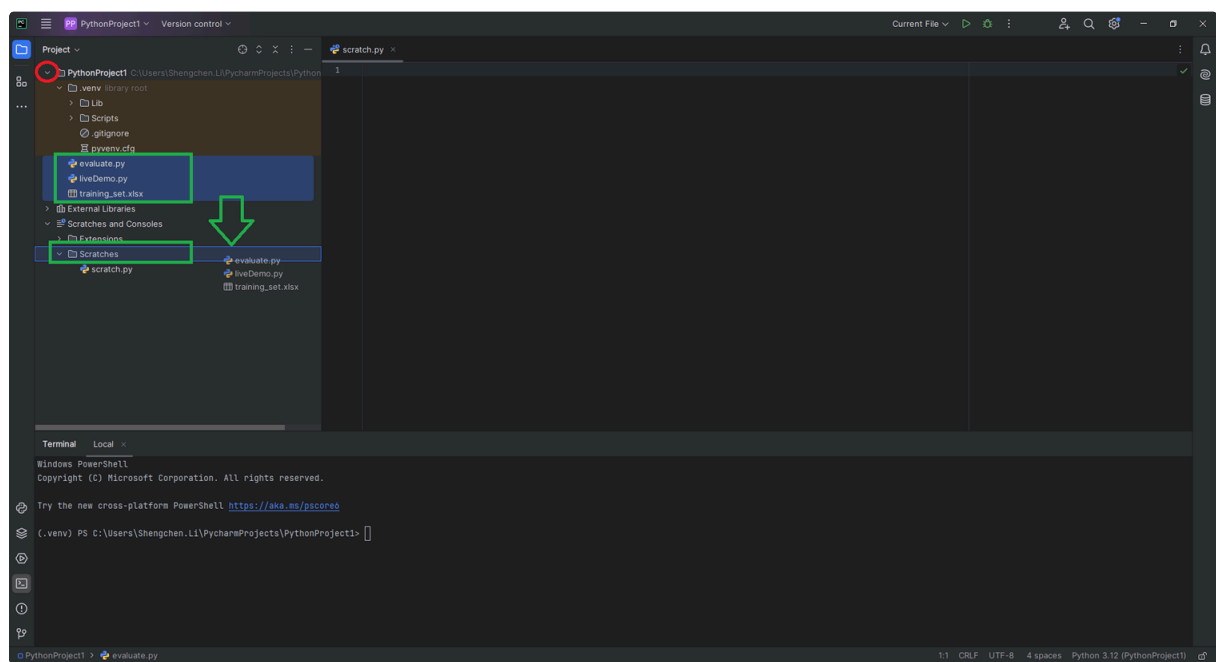
where "xxxx" is the package name.



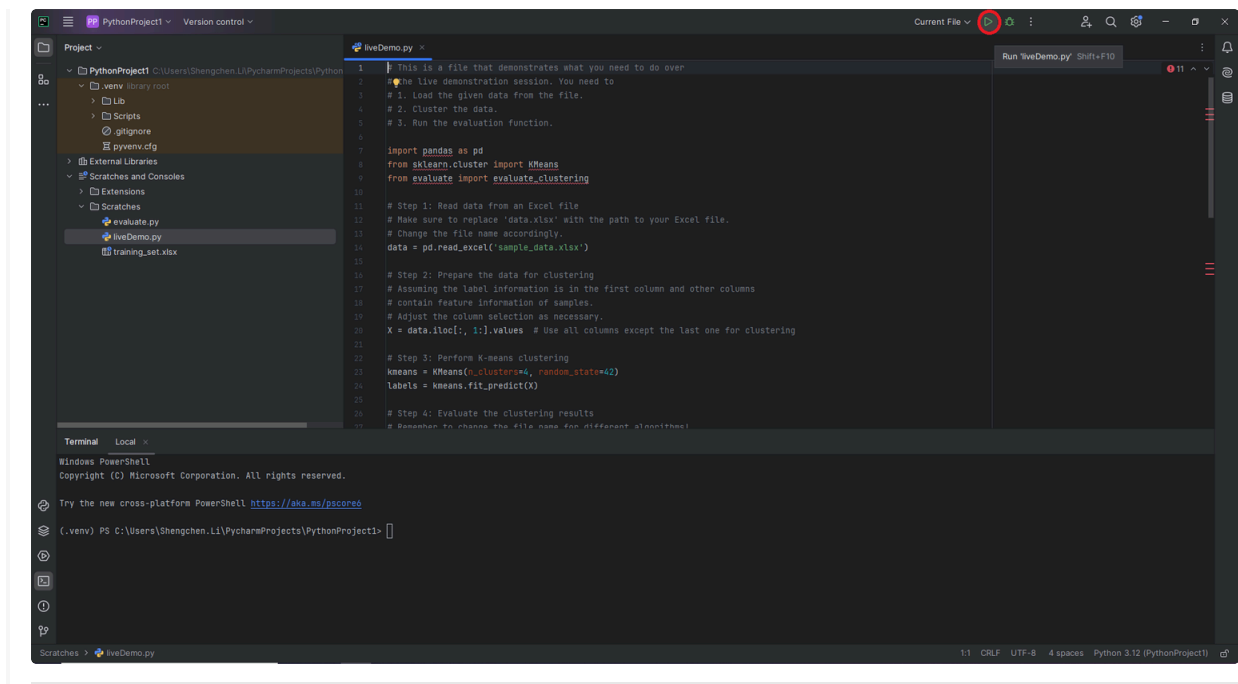
- To include existing files (such as "evaluate.py" provided by LMO), you firstly need to download the files you want and place it into the folder that the project is in



9. After storing the files, refresh the "project structure" interface by click the arrow on the left (within the red circle) twice. Then drag the files you want to move and place it into the dedicated folder you want.



10. Then you could now run the code you have written by click the button on the top-right corner.



Coursework Progress

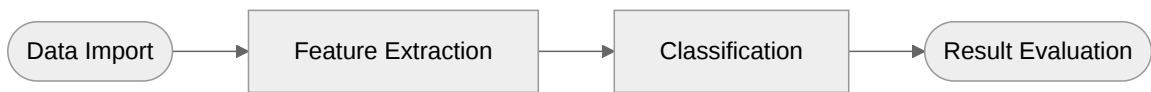
- Coursework 1: Lab Report
 - Lab: Feature Observation (Week 5–6)
 - Lab: Unsupervised Learning (Week 6, Week 8)
 - Lab: Supervised Learning (Week 9)
- Coursework 2: Unsupervised Learning Exercise
 - Lab: Unsupervised Learning (Week 6, Week 8)
 - Live Demonstration: Week 12
- Coursework 3: Supervised Learning Exercise
 - Lab: Supervised Learning (Week 9)
 - Live Demonstration: Week 13

Coursework Content

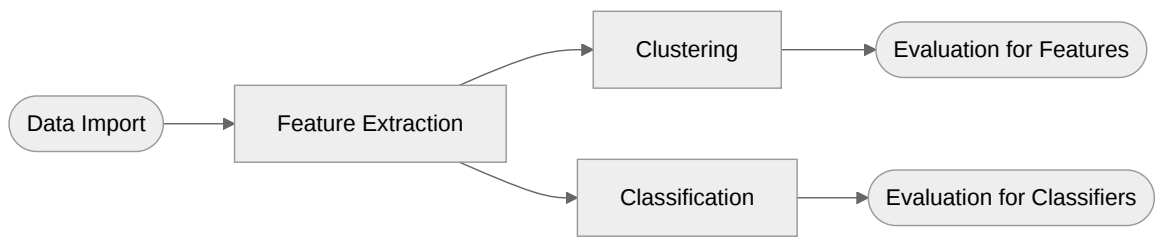
- Unsupervised Learning



- Supervised Learning



- Lab Report



Requirement for Coursework

- Understand the process of establishing an AI system
- Being able to build an AI system

i Your work will be majorly marked by what you do, *not what you have achieved*

- Finish all required tasks in the task-sheet will ensure that you could pass this module with a reasonably high score.

Lab 1 (Feature Observation) Aims

- Get familiar with Python syntax
- Being able to find features of data
- Being able to draw diagrams to demonstrate the distribution of features
- Being able to use Chatbot for programming

Lab 2 (Unsupervised Learning) Aims

- Being able to find three sets of features for data
- Being able to cluster the features with:
 - k-means
 - GMM
 - hierarchical clustering

Lab 3 (Supervised Learning) Aims

- Being able to find three sets of features for data
- Being able to classify the data with:
 - kNN
 - Naive Bayes
 - Decision Tree
- Being able to use ensemble classifiers

- Random Forest
- (Advanced) ensemble multiple simple classifiers
- Evaluate classifier with cross-validation

References

Notebooks +

Feature Selection

Instruction of Live Dem...

Supervised Learning 

Unsupervised Learning

- There are four Python scripts available for your reference.
- You could reuse the code in the script to implement the AI system for your coursework.



If you could reuse the code to perform the coursework tasks, you will be OK in this module

Sample Questions

- There are a set of sample questions on LMO, which contains
 - a set of MCQs (18 questions)
 - 5 computation questions
 - programming questions
 - ◆ fill in the blanks
 - ◆ writing code

Exam Questions

- The final exam is composited by
 - a set of MCQs (18)
 - 2 computation questions
 - 2 fill-in-the-blank programming questions
- The resit exam is composited by
 - a set of MCQs (18)
 - 3 computation questions
 - 2 Python-script writing questions